MIL-B-0036408A (DSA-DM)
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MIL-B-36408 (DM)
10 March 1965

MILITARY SPECIFICATION

BOX, PATHOLOGICAL SPECIMEN SHIPPING, INSULATED, Plastic

This limited coordination Military Specification has been prepared by the Defense Personnel Support Center based upon currently available technical information, but has not been approved for promulgation as a coordinated revision of Military Specification MIL-B-36408(DM). It is subject to modification. However, pending its promulgation as a coordinated Military Specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers one type of plastic insulated box, suitable for storage and shipment of frozen or refrigerated medical specimens and biologicals.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

V-T-276	Thread, Cotton.
PPP-B-636	Box, Fiberboard.
PPP-F-320	Fiberboard, Corrugated, and Solid, Sheet Stock (Container Grade), and Cut Shapes.
MIL-C-496	Clip, End, Strap.
MIL-W-530	Webbing, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-B-543	Buckles, Tongueless and Web Strap.

STANDARDS

Federal

FED-STD-751	Stitch, Seam, and Stitching.

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center

20759	Box, Pathological Specimen Shipping, Insulated, Plastic, 2 cu ft.

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Insulation. The insulation for the box and lid shall be fabricated from molded expanded polystyrene - foam plastic. The polystyrene - foam shall have a density of 2-1/2 ± 1/4 pounds per cubic foot, and a maximum thermal conductivity (K factor) of 0.23 B.T.U./ sq. ft./ hr./ °F./in. -- at a temperature of 75° Fahrenheit. The insulation for the box shall be acid-and alkaline-resistant.

3.1.2 Fiberboard. The fiberboard for the outer box shall be a water-resistant class, solid, fiberboard laminate with an outside off-white marble color having an overall minimum thickness of 0.075 inch and shall conform to type SF, grade W5s of PPP-F-320. Laminates comprising the overall solid board shall be bonded with a water-resistant adhesive resin, and the final fiberboard product shall have a basis weight of 310 ± 5 pounds. The weather-proof-treated fiberboard shall be moisture - and water-resistant and shall not delaminate.

3.1.3 Carrying strap harness.

3.1.3.1 Webbing. Webbing for the carrying strap harness shall be black in color, and shall conform to MIL-W-530 for type IIa, or type III, class 4, 1-1/2 inches wide, except that the maximum amount of copper-8-quinolinolate allowed shall be 1.28 percent.

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3.1.3.2 Buckle. Buckles for the straps shall conform to either of the following requirements of MIL-B-543:

a. Type II, Style 4, Class 1, and Size 1-1/2 inches.

b. Type III, Style 3 (construction "A"), Class 1, and Size 1-1/2 inches.

3.1.3.3 End clips. End clips for the strap shall conform to MIL-C-496 for type I, class 1, size 1-1/2 inches.

3.1.3.4 Thread. Thread for securing the harness strap webbing shall conform to V-T-276 for type IB3.

3.2 Construction. The pathological specimen shipping box shall consist of an insulated container encased with a laminate fiberboard, with a two-handled carrying strap. The components shall be fabricated and assembled in accordance with Drawing 20759. The assembled specimen shipping b ox shall have a weight (empty) of 12 to 14 pounds.

3.2.1 Insulated container. The insulated container shall consist of a rectangular body and a detachable plugged lid. Both components shall be of one-piece molded construction. All inside and outside surfaces of the body and lid shall have a molded skin surface finish. When the lid is secured to the body, the plug section shall form a snug fit with the inside walls of the body. The container in the upright position shall be leakproof, and shall show no evidence of leakage when tested per 4.4.1. The container shall have a minimum load capacity of 75 pounds.

3.2.2 Fiberboard box. The fiberboard shall encase the insulated container as indicated in Drawing 20759. The inside bottom of the laminated fiberboard case shall be chemically bonded to the outside bottom of the foam-plastic box with a bonding agent (adhesive) compatible with the foam-plastic and fiberboard without deleterious effects on either material. The lid shall extend over the sides of the box so as to make a close fit with the top of the fiberboard case without undue binding. Joints shall be firmly secured with staples suitable for the intended purpose, as shown on the drawing.

3.2.3 Carrying strap harness. The carrying strap harness shall consist of two straps with carrying handles. As indicated in Drawing 20759, one strap shall fit around the height of the specimen shipping box, and the other strap shall fit around the girth of the specimen shipping box. Each strap shall have a corrosion-resisting metal buckle (see 3.1.3.2) that is secured in a loop formed by the webbing of the strap. The loop shall be fastened by box tack sewing as indicated by Drawing 20759. The raw end of the loop, to prevent raveling, shall either be turned under and double stitched, or impregnated

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with a hardening material. The opposite end of the strap shall terminate with an end clip specified in 3.1.3.3. The webbing shall be inserted the full depth of the end clip and shall be flat at the point of entry. The end clip shall be firmly attached without cutting the webbing. The straps shall be attached to each other as indicated in Drawing 20759, to provide a one-piece sling harness. Handle configurations, made of the same webbing material, shall be attached at points above and below the junction points of the straps. Attachment points of the sling strap harness (strap cross points, handle configurations, and buckle-securing points) shall be accomplished by box tack sewing. The thread shall conform to 3.1.3.4. All stitching shall conform to stitch type 301 of Federal Standard No. 751, with six to eight stitches per inch. Thread tension shall be maintained so that there is no loose stitching, and the lock shall be embedded in the webbing material. The straps shall be of suitable length so that the loose ends come together at the top of the closed container and around the girth of the container, with the buckles located approximately halfway across the cover of the closed container and approximately halfway across the girth of the container. The clip ends of the buckled straps shall extend through the buckles with a minimum gripping length of six inches.

3.3 Identification marking. The name or registered trademark of the manufacturer shall be permanently and legibly marked on the exterior surface of the bottom of each fiberboard box. Decalcomanias shall not be employed.

3.4 Workmanship. Workmanship shall be first class throughout. The boxes and harness straps shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturers of the fiberboard, webbing, buckles, end clips, and thread, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of inspection shall be the assembled specimen shipping box with carrying strap harness.

Table I. For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II.

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Table II. For sampling

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level	AQL
Plastic box only	Leakage	3.3.1	4.4.1	S-2	*

*Acceptance number shall be zero.

4.3 Examination. The unit box shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major

101 Plastic box and lid not free of chips, cuts, and indentations.
102 Plastic lid does not fit snugly in foam box.
103 Fiberboard box and lid not free of deformation.
104 Fiberboard box and lid not free of tears, cuts, holes, or indentations.
105 Sewing on strap harness not free of loose stitching.
106 Lid of fiberboard box does not fit closely to top of box without undue binding.
107 Plastic body and lid not free of crushed edges or gouges.
108 Fiberboard box or lid not free of crushed, ragged, or uneven edges.
109 Stapled sides (corners) of fiberboard body or lid not properly secured (stapling not well-clinched or does not pass through all pieces to be fastened).
110 Webbing straps not free of cuts or holes.
111 Assembled box on strap harness not free of missing component.
112 Clip not adequately or securely attached to webbing of strap.

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Categories and defects* (cont'd.)

Minor

201 Identification marking not complete, not permanent, illegible, or missing.
202 Fiberboard box or lid not free of chipping at edges.
203 Fiberboard box lid not free of scuff or deep scratches.
204 Plastic box not free of foreign matter or loose or ingrained dirt.
205 Fastening staples not free of visible cracking or rust.
206 Webbing not flat at point of entry to clip.

*Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The unit box shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Leakage. The plastic box shall be set on newsprint or similar type paper and filled with tap water to within 3 inches from the top of box. The water-filled box shall be allowed to stand for 24 hours before examining for evidence of leakage on the paper and for moisture on the sides of the box.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One box, as specified, constitutes one unit of issue.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. One unit shall be overpacked in a fiberboard box of appropriate size constructed in accordance with PPP-B-636, type CF, class weather-resistant. Closure of each fiberboard box shall be [illegible] as specified in the appendix, and waterproofing shall conform to 30.4 of PPP-B-636.

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5.2.2 Level B.

5.2.2.1 Exterior container. One unit shall be overpacked in a fiberboard box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic, and method II closure shall be utilized.

5.2.3 Level C. The commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable level of packaging and packing (see 5.1 and 5.2).
c. Federal stock number (FSN).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

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6.3 This specification covers the following item listed in the Federal Supply Catalog:

Federal Stock No.	Item Identification
8115-226-1199	BOX, PATHOLOGICAL SPECIMEN, SHIPPING, INSULATED, Plastic, 2 cu. ft.

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 8115-0340

Review activities:

Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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